FIFTH ELEMENT

FUNCTIONAL SPECIFICATION DOCUMENT — PROTOTYPE 1

OBJECTIVE

Prototype 1 aims to demonstrate the concept behind the Fifth Element smart garden: a system capable of remotely controlling watering and lighting, providing useful information to the user, and enabling an initial visual assessment of the plant through a camera.
The goal is not yet to create a fully autonomous growing system. Instead, it is to build a first functional hardware and software foundation that future prototypes can build upon.

Functional scope of prototype 1



F01 - Watering Management

Mandatory

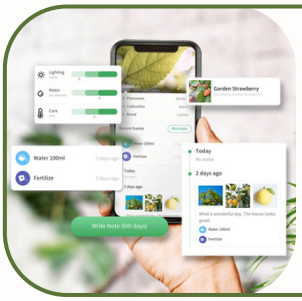
- Manual control through the application.
- Water delivery through a hydraulic circuit.
- Pump shutdown upon user command.
- Display of the pump's operating status.



F02 - Lighting Management

Mandatory

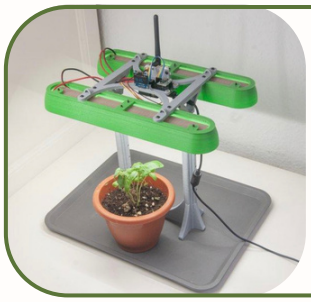
- Turn the light on and off.
- Display the current lighting status.
- Ensure reliable on/off control.
- Allow for the future : scheduled lighting.



F03 - Application and Interface

Mandatory

- Status of the pump and lighting.
- Control watering and lighting.
- Show remaining water in the reservoir.
- Access to the camera of the plant.
- Report communication errors.



F04 - Visual Monitoring

Function to Be Defined

- Capture and view images of the plant.
- Verify image quality and availability.
- Assess the feasibility of an initial visual analysis of the plant's condition.
- Design an architecture that allows image processing capabilities to be expanded in future prototypes.

Detailed Functional Requirements

Reference	Requirement	Acceptance criterion
EX-01	The user can remotely activate the pump.	The pump starts after receiving a valid command.
EX-02	The user can remotely stop the pump.	The pump stops after receiving a valid command.
EX-03	The system limits the pump's operating time.	The pump stops automatically when the configured maximum duration is reached.
EX-04	The user can remotely control the lighting.	The light switches on and off, and its status is displayed in the application.
EX-05	The application displays the reservoir water level.	A water-level measurement is available and updated.
EX-06	The application displays equipment status.	Displayed statuses match the actual system states, and outdated data is identified.
EX-07	The user can observe the plant through a camera.	A usable image can be captured and displayed.
EX-08	The system supports future visual analysis.	The image data can be accessed by the intended image-processing system.
EX-09	The system reports significant errors.	Communication loss or system unavailability is indicated in the application.